Experiment No. 6

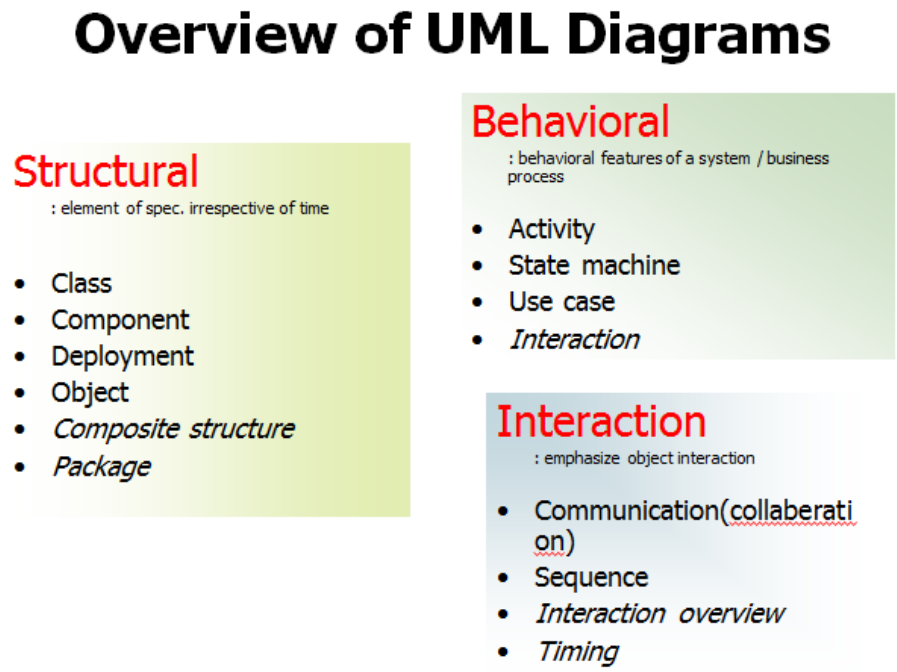
Title: Creation of Structural UML Diagrams

Batch: B3 **Roll No.: 16014225801** **Experiment No.: 6**

Aim: Creation of UML Diagrams

Resources needed: IBM Rational Rose/Open Source UML Tool-star UML

Theory



UML specification defines two major kinds of UML diagram: **structure diagrams and behaviour diagrams. Structural diagrams** show the static structure of the system and it's parts on different abstraction and implementation levels and how they are related to each other. The elements in a structure diagram represent the meaningful concepts of a system, and may include abstract, real world and implementation concepts. **Behavioral diagrams** show the dynamic behavior of the objects in a system, which can be described as a series of change s to the system over time.

I. Structure Diagrams:

Structure diagram shows static structure of the system and its parts on different abstraction and implementation levels and how those parts are related to each other. The elements in a structure diagram represent the meaningful concepts of a system, and may include abstract, real world and implementation concepts.

Structure diagrams are not utilizing time related concepts; do not show the details of dynamic behavior. However, they may show relationships to the behaviors of the classifiers exhibited in the structure diagrams.

1. Class diagram is static structure diagram describing structure of a system on the (lowest) level of classifiers (classes, interfaces, etc.). It shows system's classifiers, their attributes, and the relationships between classifiers.

2.Object diagram shows instances of classifiers and links (instances of associations) between them.

3. Package Diagram shows packages and dependencies between the packages. Models allow to show different views of a system, for example, as multi-layered (aka multi-tiered) application

4. Component Diagram shows components and dependencies between them. This type of diagrams is used for Component-Based Development (CBD), to describe systems with Service-Oriented Architecture (SOA).

Composite Structure diagram could be used to show:
Internal structure of a classifier, Internal Structure diagrams show internal structure of a classifier - a decomposition of the classifier into its properties, parts and relationships, Behaviour of collaboration, Collaboration use diagram shows objects in a system cooperating with each other to produce some behavior of the system.

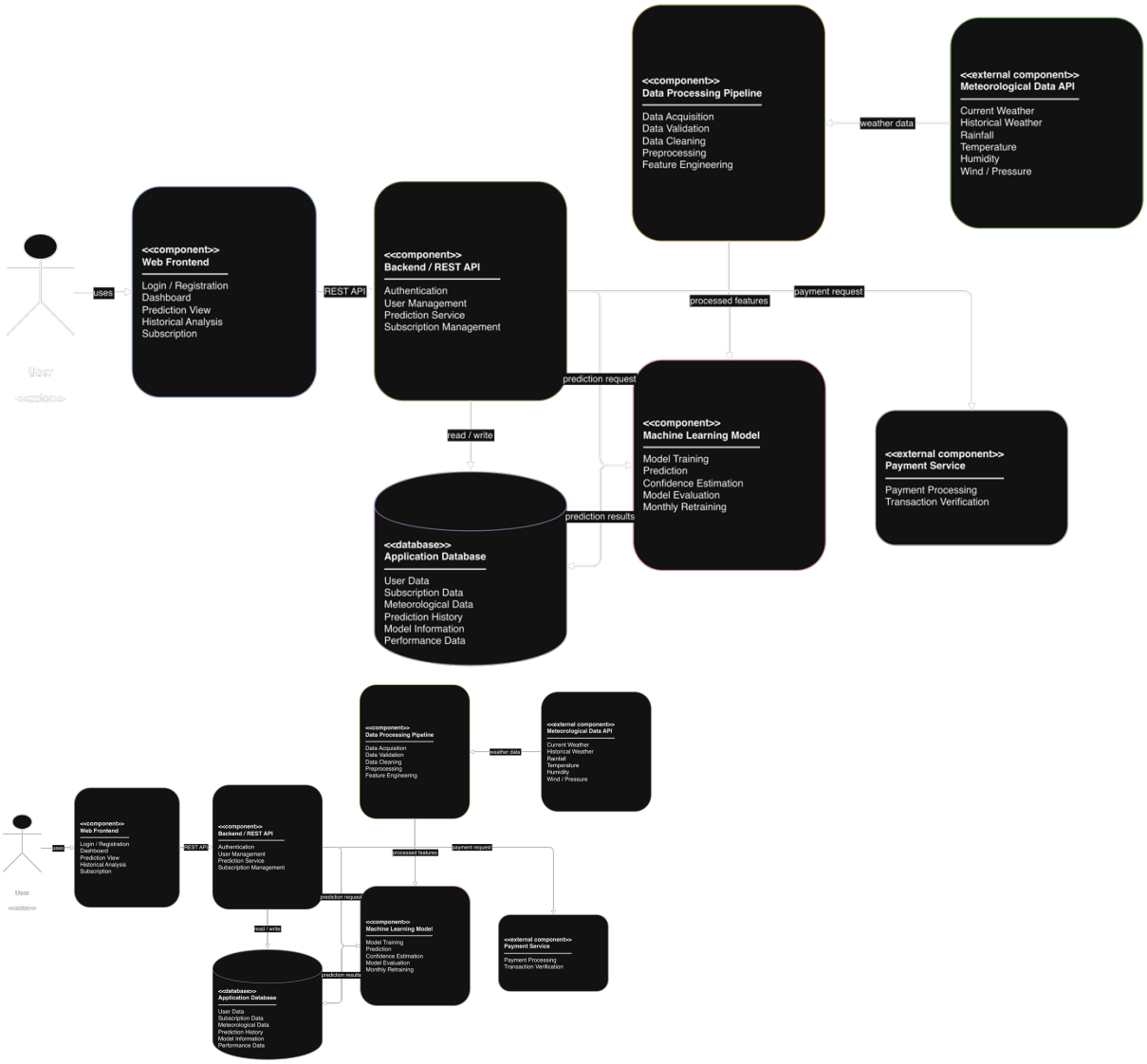
Deployment diagram shows execution architecture of a system that represents the assignment (deployment) of software artifacts to deployment targets (usually nodes).

Profile diagram is auxiliary UML diagram which allows defining custom stereotypes, tagged values, and constraints. The Profile mechanism has been defined in UML for providing a lightweight extension mechanism to the UML standard. Profiles allow to adapt the UML meta model for different platforms (such as J2EE or .NET), or domains (such as real-time or business process modeling).
Profile diagrams were first introduced in UML 2.0.

Procedure:
Prepare mentioned diagrams for chosen problem using Rational Rose/ any other Open Source UML tool.

Results:

Fig. Class Diagram



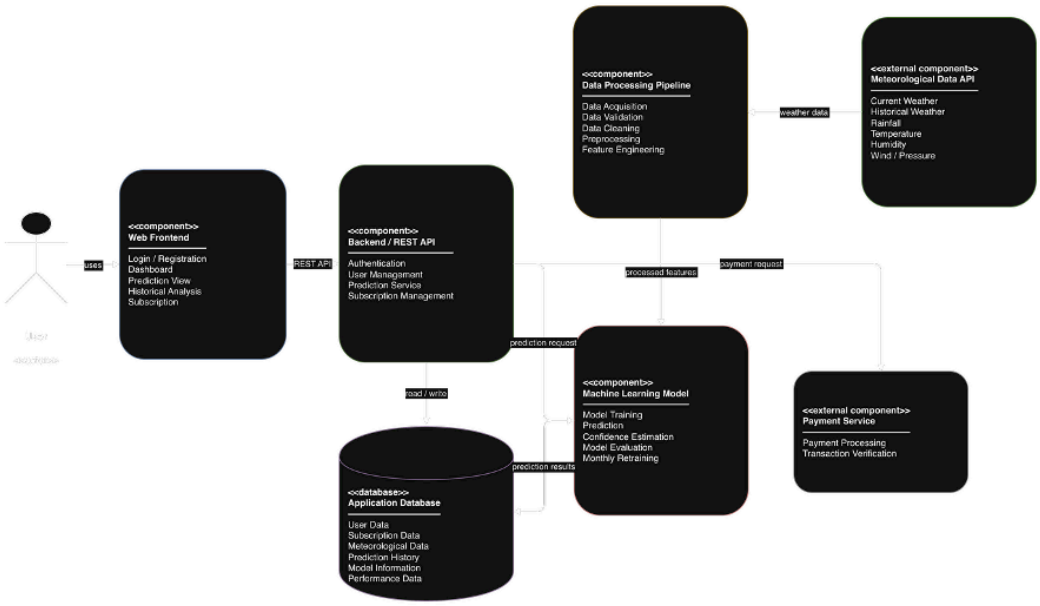


Fig. Component Diagram



Fig. Deployment Diagram

Questions:

1. Explain the difference between component diagram and deployment diagram
2. Give example of deployment diagram for modelling a fully distributed system.

Answers:

1. Component Diagram vs Deployment Diagram: A component diagram shows the software components of a system and how they depend on or interact with each other. It focuses on the logical structure of the software, such as modules, services, databases, and interfaces. A deployment diagram shows the physical arrangement of the system, including servers, computers, devices, and network connections, and explains where software components are deployed. In simple terms, a component diagram shows "what software is present and how it connects," while a deployment diagram shows "where the software runs physically."
2. Example of a Deployment Diagram for a Fully Distributed System: An example is an online banking system where different services are distributed across multiple servers. The client application runs on users' computers or mobile phones and connects through the internet to a web server. The web server communicates with separate application servers handling authentication, transactions, and account management. These servers connect to distributed database servers that store customer and transaction data. Each node performs a specific function and communicates with other nodes through a network, making the system fully distributed.

Conclusion:

Hence we made the diagram and completed the diagrams required.

Grade: AA / AB / BB / BC / CC / CD /DD

Signature of faculty in-charge with date

References:

Books/ Website:

1.Michael Blaha, James Rumbaugh, "Object-Oriented Modeling and Design with UML", Prentice-Hall of India, 2nd Edition

2. Mahesh P. Matha, "Object-Oriented Analysis and Design using UML", Prentice-Hall of India

3. Timothy C Lethbridge, Robert Laganiere, "Object-Oriented Software Engineering – A practical software development using UML and Java", Tata McGraw-Hill, New Delhi.

4. http://www.uml-diagrams.org/uml-23-diagrams.html